Note: to ensure compatibility with multiboard designs, true 3v3 rail (referenced to ground) must be provided on the subsequent board to provide ground reference to hall effect sensors and mosfet drivers.

Ground rail also to be provided to subsequent board to ensure common references for control signals

Alternatively, run 1 nucleo per board

H1
MountingHole

H2
MountingHole

H3
MountingHole

H4
MountingHole

Voltage Sensing
CS1
CS2
CS3
CS1 Select
CS2 Select
CS3 Select
Floating 3v3
SCLKBus
MOSI Bus
MISO Bus
Floating 3v3
SPI SCLK
SPI MOSI
SPI MISO
File: V_Sense.kicad_sch

String output

PV1+
XT60_1
PV Connector 1

Shuffle Converter 1
Input_T
Output
Input_B
Floating 3v3
Gate Driver 3v3
I_Sense_Out_Aph
I_Sense_Out_Bph
PWM_A_H
PWM_A_L
PWM_B_H
PWM_B_L
S1_A_H
S1_A_L
S1_B_H
S1_B_L
ILA_1
ILB_1
File: Shuffle_Converter_Top.kicad_sch

PV2+
XT60_2
PV Connector 2

XT60_4
PV Connector 4

Gate Driver 3v3
Floating 3v3

Shuffle Converter 2
S2_A_H
S2_A_L
S2_B_H
S2_B_L
ILA_2
ILB_2
PWM_A_H
PWM_A_L
PWM_B_H
PWM_B_L
I_Sense_Out_Aph
I_Sense_Out_Bph
Input_T
Output
Input_B
Gate Driver 3V3
LDO 3v3
File: Shuffle_Converter.kicad_sch

PV3+
XT60_3
PV Connector 3

TP12 TestPoint
TP14 TestPoint
TP3 TestPoint
TP1 TestPoint
TP11 TestPoint
TP13 TestPoint
TP4 TestPoint
TP2 TestPoint
LDO 3v3
Gate Driver 3v3
GND

Connections to be changed, confirm new pinout arrangment with less gate drive signals but new SPI integration

U1A
NUCLEO-G474RE
PC10
PC12
VDD
PB8-BOOT0
NC_CN7_9
NC_CN7_11
PA13
PA14
PA15
GND_CN7_19
PB7
PC13
PC14
PC15
PF0
PF1
VBAT
PC2
PC3
PC11
PD2
E5V
GND_CN7_8
NC_CN7_10
IOREF
RST
+3V3
+5V
GND_CN7_20
GND_CN7_22
VIN
NC_CN7_26
PA0
PA1
PA4
PB0
PC1/PB9
PC0/PA15
ILA_1
ILA_2
ILB_1
ILB_2

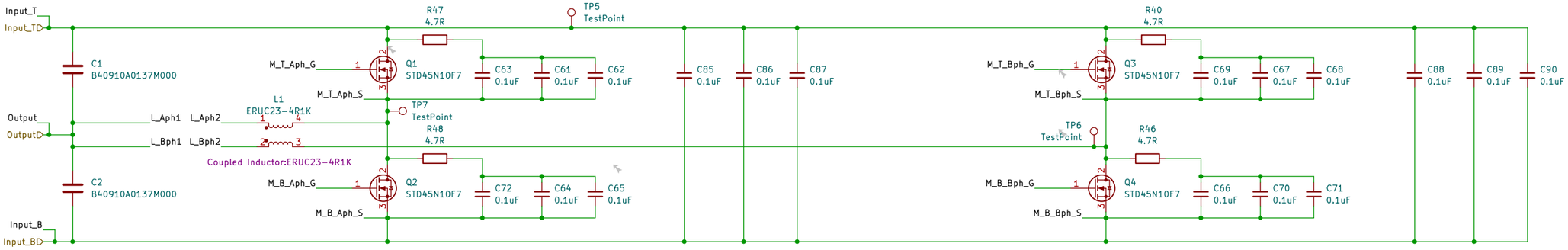
GND
1k R26
LDO 3v3
680 R25
D4 LED_G
GND

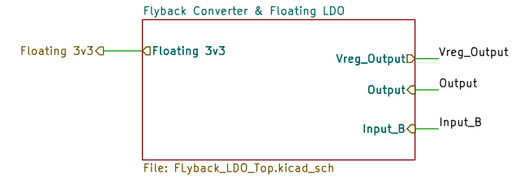
S2
L203011MS02Q

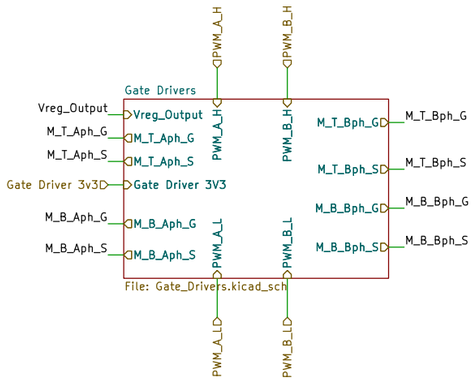
R23 680
D1 LED_R
+3V3
GND
100 R24
D2 LED_B
GND

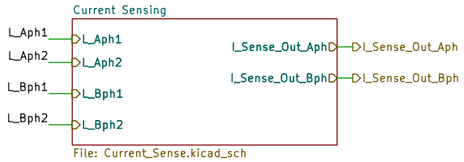
U1B
NUCLEO-G474RE
PC9
PB8
PB9
AVDD
GND_CN10_9
PA5
PA6
PA7
PB6
PC7
PA9
PA8
PB10
PB4
PB5
PB3
PA10
PA2/PC4
PA3/PC5
PC8
PC6
PC5
5V_USB_CHGR
NC_CN10_10
PA12
PA11
PB12
PB11
GND_CN10_20
PB2
PB1
PB15
PB14
PB13
AGND
PC4
NC_CN10_36
NC_CN10_38
CS1
CS2
CS3
S1_A_L
S1_A_H
SPI MiSO
SPI MOSI
SPI SCLK
S1_B_H
S1_B_L
S2_A_H
S2_B_L
S2_B_H
S2_A_L

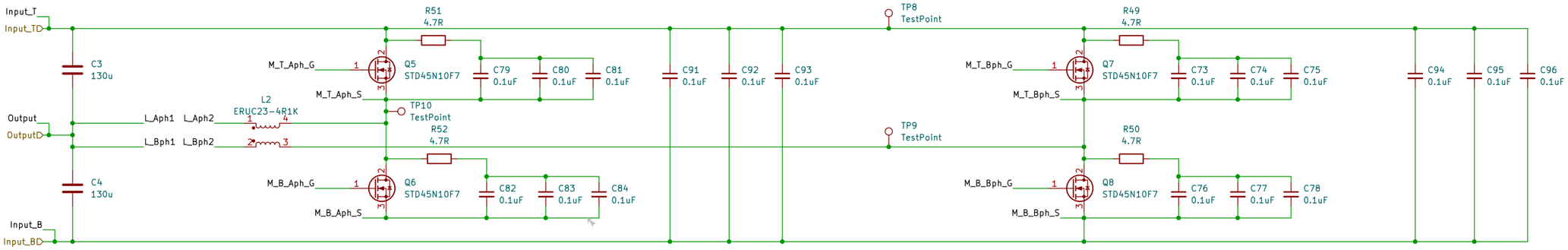
+3V3
SW1
L101011MS02Q
Gate Driver 3v3


Input_T
Input_T
C1
B40910A0137M000
Output
Output
L_Aph1
L_Aph2
L_Bph1
L_Bph2
L1
ERUC23-4R1K
Coupled Inductor:ERUC23-4R1K
C2
B40910A0137M000
Input_B
Input_B
M_T_Aph_G
M_T_Aph_S
M_B_Aph_G
M_B_Aph_S
Q1
STD45N10F7
Q2
STD45N10F7
R47
4.7R
R48
4.7R
TP5
TestPoint
TP7
TestPoint
C63
0.1uF
C61
0.1uF
C62
0.1uF
C72
0.1uF
C64
0.1uF
C65
0.1uF
C85
0.1uF
C86
0.1uF
C87
0.1uF
M_T_Bph_G
M_T_Bph_S
M_B_Bph_G
M_B_Bph_S
Q3
STD45N10F7
Q4
STD45N10F7
R40
4.7R
R46
4.7R
TP6
TestPoint
C69
0.1uF
C67
0.1uF
C68
0.1uF
C66
0.1uF
C70
0.1uF
C71
0.1uF
C88
0.1uF
C89
0.1uF
C90
0.1uF

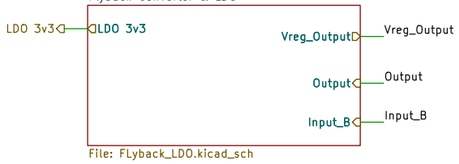

Flyback Converter & Floating LDO
Floating 3v3
Floating 3v3
Vreg_Output
Vreg_Output
Output
Output
Input_B
Input_B
File: FLyback_LDO_Top.kicad_sch

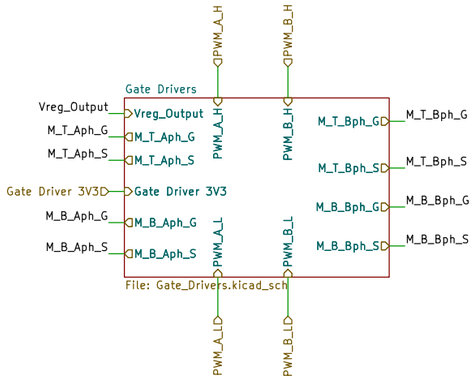

Gate Drivers
PWM_A_H
PWM_B_H
Vreg_Output
Vreg_Output
M_T_Aph_G
M_T_Aph_S
Gate Driver 3v3
Gate Driver 3V3
M_B_Aph_G
M_B_Aph_S
PWM_A_H
PWM_B_H
PWM_A_L
PWM_B_L
M_T_Bph_G
M_T_Bph_S
M_B_Bph_G
M_B_Bph_S
M_T_Bph_G
M_T_Bph_S
M_B_Bph_G
M_B_Bph_S
File: Gate_Drivers.kicad_sch
PWM_A_L
PWM_B_L

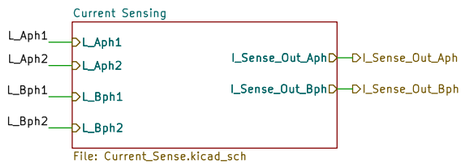

Current Sensing
I_Aph1
I_Aph2
I_Bph1
I_Bph2
I_Aph1
I_Aph2
I_Bph1
I_Bph2
I_Sense_Out_Aph
I_Sense_Out_Bph
I_Sense_Out_Aph
I_Sense_Out_Bph
File: Current_Sense.kicad_sch

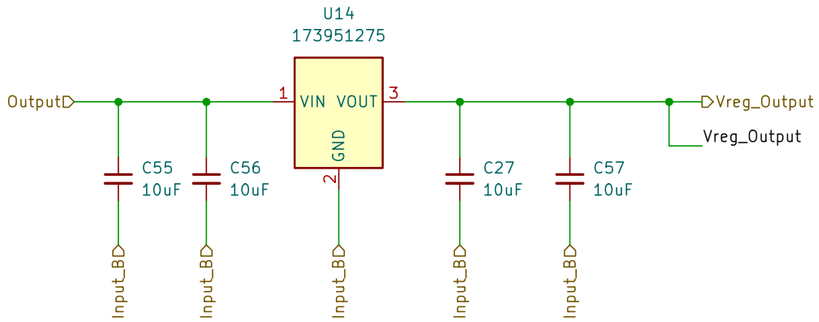


Flyback Converter & LDO



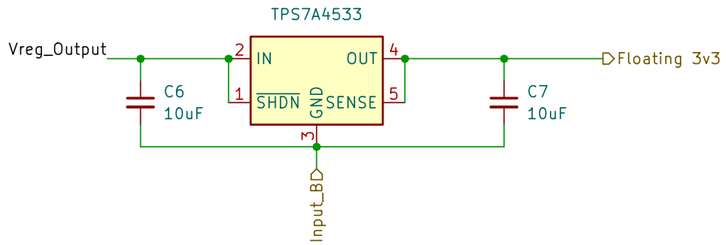


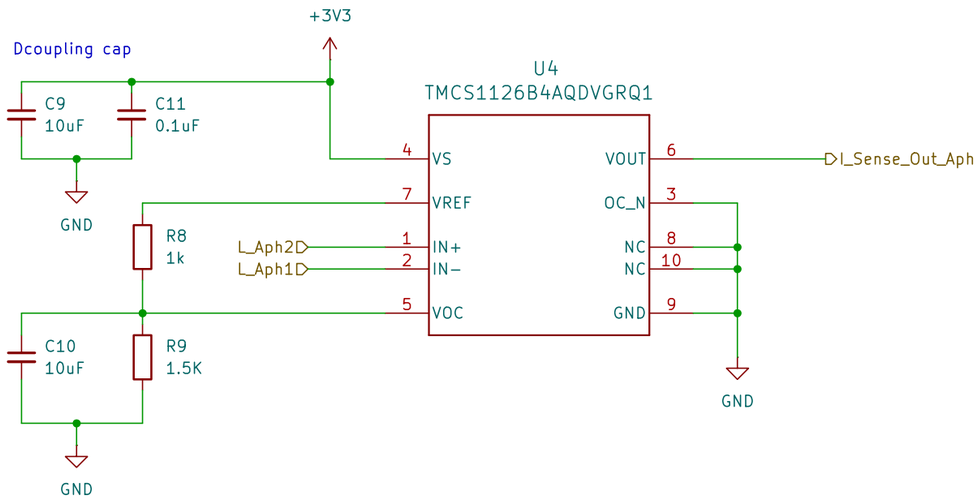


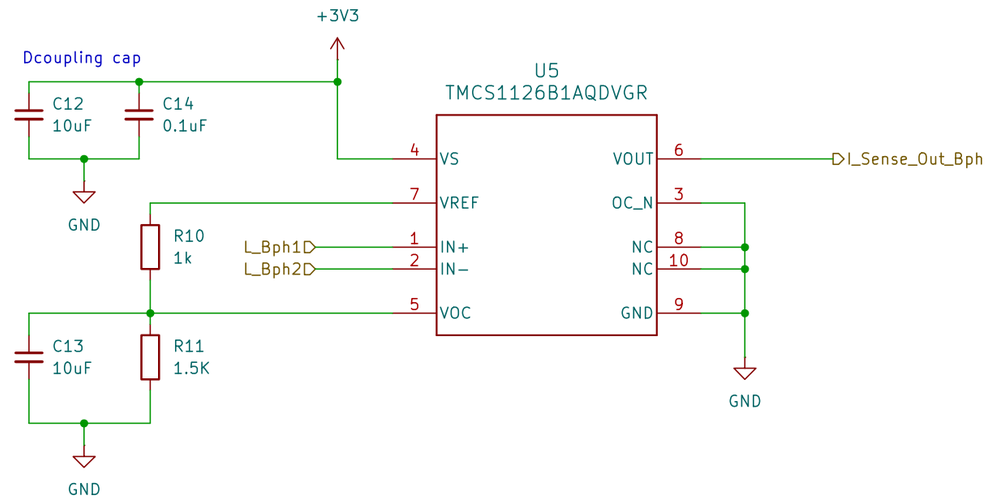


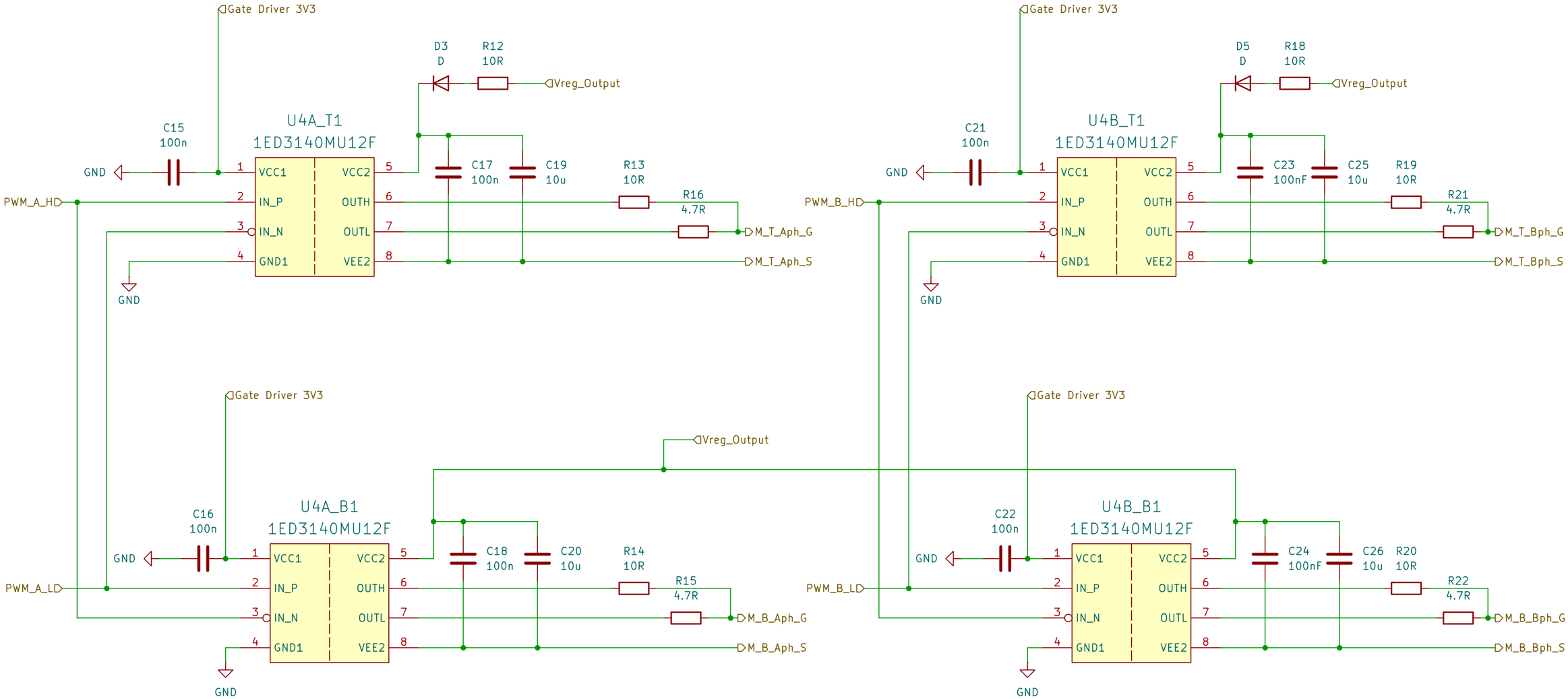
Floating 3v3 LDO

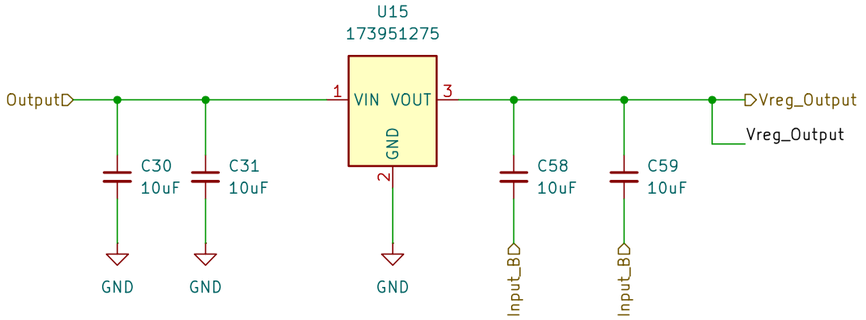
U2

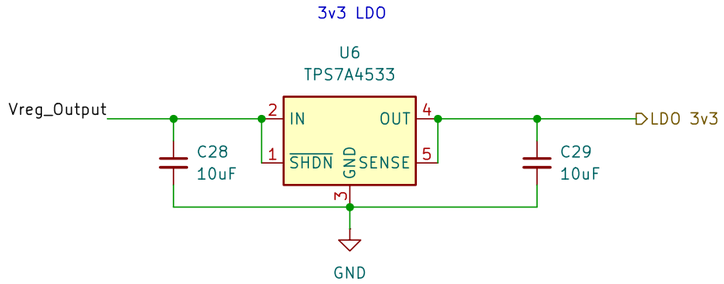


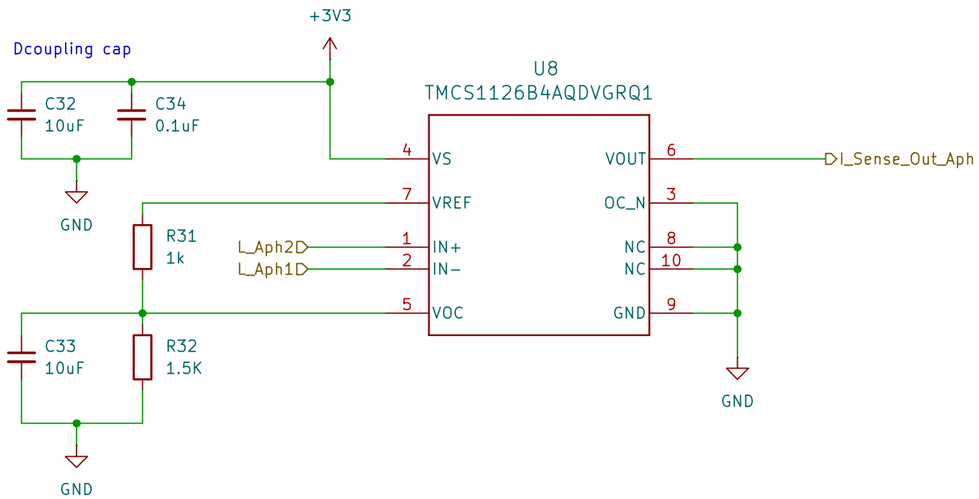

+3V3
Dcoupling cap
C9
10uF
C11
0.1uF
GND
U4
TMCS1126B4AQDVGRQ1
4
VS
7
VREF
1
IN+
2
IN−
5
VOC
VOUT
6
OC_N
3
NC
8
NC
10
GND
9
I_Sense_Out_Aph
R8
1k
L_Aph2
L_Aph1
C10
10uF
R9
1.5K
GND
GND

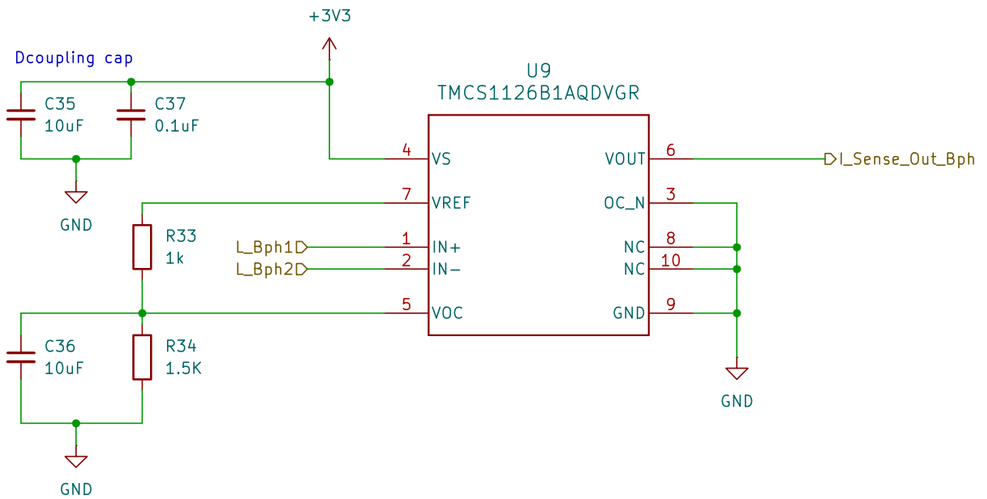

+3V3
Dcoupling cap
C12
10uF
C14
0.1uF
GND
U5
TMCS1126B1AQDVGR
4
VS
7
VREF
1
IN+
2
IN−
5
VOC
VOUT
6
OC_N
3
NC
8
NC
10
GND
9
I_Sense_Out_Bph
R10
1k
L_Bph1
L_Bph2
C13
10uF
R11
1.5K
GND
GND


Gate Driver 3V3
D3
D
R12
10R
Vreg_Output
C15
100n
U4A_T1
1ED3140MU12F
GND
PWM_A_H
1
VCC1
2
IN_P
3
IN_N
4
GND1
VCC2
5
OUTH
6
OUTL
7
VEE2
8
C17
100n
C19
10u
R13
10R
R16
4.7R
M_T_Aph_G
M_T_Aph_S
GND
Gate Driver 3V3
D5
D
R18
10R
Vreg_Output
C21
100n
U4B_T1
1ED3140MU12F
GND
1
VCC1
2
IN_P
3
IN_N
4
GND1
VCC2
5
OUTH
6
OUTL
7
VEE2
8
C23
100nF
C25
10u
R19
10R
R21
4.7R
M_T_Bph_G
M_T_Bph_S
GND
Gate Driver 3V3
Vreg_Output
C16
100n
U4A_B1
1ED3140MU12F
GND
PWM_A_LD
1
VCC1
2
IN_P
3
IN_N
4
GND1
VCC2
5
OUTH
6
OUTL
7
VEE2
8
C18
100n
C20
10u
R14
10R
R15
4.7R
M_B_Aph_G
M_B_Aph_S
GND
Gate Driver 3V3
C22
100n
U4B_B1
1ED3140MU12F
GND
PWM_B_LD
1
VCC1
2
IN_P
3
IN_N
4
GND1
VCC2
5
OUTH
6
OUTL
7
VEE2
8
C24
100nF
C26
10u
R20
10R
R22
4.7R
M_B_Bph_G
M_B_Bph_S
GND

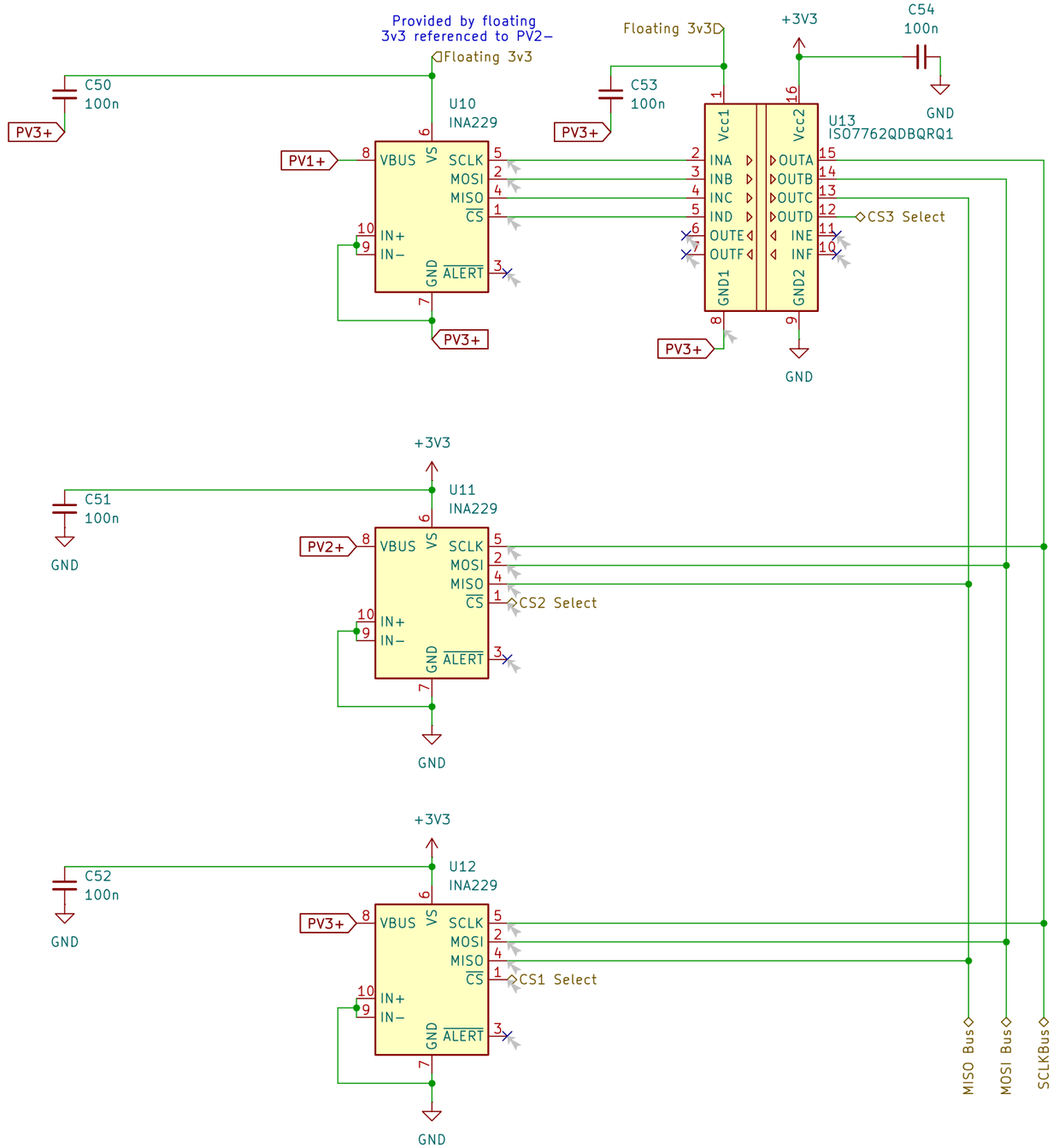

U15
173951275
Output
1
VIN
VOUT
3
GND
2
Vreg_Output
Vreg_Output
C30
10uF
C31
10uF
C58
10uF
C59
10uF
GND
GND
GND
Input_BD
Input_BD


3v3 LDO
U6
TPS7A4533
Vreg_Output
2
IN
OUT
4
LDO 3v3
1
SHDN
GND
SENSE
5
3
C28
10uF
C29
10uF
GND


+3V3
Dcoupling cap
U8
TMCS1126B4AQDVGRQ1
C32
10uF
C34
0.1uF
GND
R31
1k
L_Aph2
L_Aph1
C33
10uF
R32
1.5K
GND
4
VS
7
VREF
1
IN+
2
IN−
5
VOC
VOUT
6
OC_N
3
NC
8
NC
10
GND
9
GND
I_Sense_Out_Aph


+3V3
Dcoupling cap
U9
TMCS1126B1AQDVGR
C35
10uF
C37
0.1uF
GND
R33
1k
L_Bph1
L_Bph2
C36
10uF
R34
1.5K
GND
4
VS
7
VREF
1
IN+
2
IN−
5
VOC
VOUT
6
OC_N
3
NC
8
NC
10
GND
9
GND
I_Sense_Out_Bph

Gate Driver 3V3
D8
D
R35
10R
Vreg_Output
C38
100n
U4A_T2
1ED3140MU12F
GND
C40
100n
C42
10u
R36
10R
R39
4.7R
PWM_A_H
1
VCC1
2
IN_P
3
IN_N
4
GND1
5
VCC2
6
OUTH
7
OUTL
8
VEE2
M_T_Aph_G
M_T_Aph_S
GND
Gate Driver 3V3
D10
D
R41
10R
Vreg_Output
C44
100n
U4B_T2
1ED3140MU12F
GND
C46
100nF
C48
10u
R42
10R
R44
4.7R
PWM_B_H
1
VCC1
2
IN_P
3
IN_N
4
GND1
5
VCC2
6
OUTH
7
OUTL
8
VEE2
M_T_Bph_G
M_T_Bph_S
GND
Gate Driver 3V3
Vreg_Output
C39
100n
U4A_B2
1ED3140MU12F
GND
C41
100n
C43
10u
R37
10R
R38
4.7R
PWM_A_LD
1
VCC1
2
IN_P
3
IN_N
4
GND1
5
VCC2
6
OUTH
7
OUTL
8
VEE2
M_B_Aph_G
M_B_Aph_S
GND
Gate Driver 3V3
C45
100n
U4B_B2
1ED3140MU12F
GND
C47
100nF
C49
10u
R43
10R
R45
4.7R
PWM_B_LD
1
VCC1
2
IN_P
3
IN_N
4
GND1
5
VCC2
6
OUTH
7
OUTL
8
VEE2
M_B_Bph_G
M_B_Bph_S
GND


Provided by floating
3v3 referenced to PV2−
Floating 3v3
C50
100n
PV3+
U10
INA229
PV1+
VBUS
VS
SCLK
MOSI
MISO
CS
IN+
IN−
GND
ALERT
PV3+
Floating 3v3
C53
100n
PV3+
+3V3
C54
100n
GND
U13
ISO7762QDBQRQ1
Vcc1
Vcc2
INA
INB
INC
IND
OUTE
OUTF
GND1
OUTA
OUTB
OUTC
OUTD
INE
INF
GND2
CS3 Select
PV3+
GND
+3V3
C51
100n
GND
U11
INA229
PV2+
CS2 Select
GND
+3V3
C52
100n
GND
U12
INA229
PV3+
CS1 Select
GND
MISO Bus
MOSI Bus
SCLKBus